

Raga Mouni Batchu

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SUMMARY

Experienced software developer with 5+ years in designing, developing, and maintaining scalable solutions. Skilled in cloud technologies, backend development, with research experience applying GraphRAG and agentic AI techniques. Proven expertise in migrating legacy systems to microservices, enhancing performance, and leading high-impact teams. Served as Scrum Master for 3 years, delivering consistent results through agile practices.

TECHNICAL SKILLS

- Languages: JavaScript, Node JS, Perl, C++, Python
- AI/ML Frameworks: Huggingface, PyTorch, Langchain, Vertex AI, PySpark
- AI/ML Techniques: Supervised Instruction-based Fine tuning, RAG, GraphRAG, Agent Development
- Database: PostgreSQL, Document DB (Mongo DB)
- Framework: Spring boot, Kubernetes
- Cloud: AWS (S3, SQS, RDS, SSM, EC2, CloudFront, CloudWatch, Docker)
- Others: Informatica, Jenkins, Grafana, graylog, Kafka, GraphQL, Hasura, Prometheus, REST API

EDUCATION

West Chester University

Master of Science in Computer Science (CGPA:4/4)

Current

Sri Vishnu Engineering College for Women

Bachelor of Technology in Computer Science (CGPA:9/10)

05/2019

RESEARCH & PUBLICATIONS

Bidirectional Translation of ASL and English using CNN and Transformers

Published

- Built a bidirectional ASL-English translation system combining a CNN-BiLSTM sign-to-gloss model, Transformer-based gloss-to-sentence models, and a gloss-to-sign animation pipeline.

A Survey on Data Visualization Techniques for International Trade

Published

Effect of MLA and RoPE in agglutinative languages like Telugu

Draft Complete

- Built an end-to-end Telugu language modeling pipeline using morphology-aware SentencePiece tokenization and controlled three-way ablations to isolate architectural gains from RoPE and MLA.
- Achieved 20.6% lower perplexity by replacing positional embeddings with RoPE and reduced attention parameters by 31% using MLA, while demonstrating 2x higher cross-morpheme attention.

ASL recognition and game-based interaction for Deaf Learners

Draft Complete

- Developed an ASL learning game with a custom backtracking-based puzzle generator and a spatial-temporal Transformer for gloss recognition, achieving 89% Top 5 accuracy on WL-ASL dataset.

KG-RAG for Tox21: Explainable Drug Toxicity Prediction

Work in Progress

- Developing a knowledge graph-based RAG application on the toxicity domain, integrating Tox21, PubChem, DSSTox, ChemBL, UniProt and Reactome data products.

DSABench: New Benchmark for evaluating non-coding DSA problem statements

Work in Progress

- Created DSABench, a university-level benchmark of data structures and algorithms questions with curated reasoning-based answers, designed to evaluate SOTA LLMs on core CS understanding and algorithmic reasoning without relying on code execution or tool calls.

WORK EXPERIENCE

athenahealth, Bengaluru, India

Senior Member of Technical Staff

06/2023 – 08/2024

- Led the migration of call-related features from a legacy system to microservices, adding new functionalities and saving \$732k annually by reducing manual client team workloads.
- Designed and implemented a comprehensive monitoring system for messaging services, optimizing the operational visibility of the microservices.
- Created dashboards in Graylog on-demand basis, which eased the debugging of production issues

- Improved SMS opt-in rates for patients by 30% by integrating a DTMF node in call flows.
- Spearheaded a team of 5 engineers, resolving critical production issues and ensuring consistent system uptime.
- Served as Scrum Master for 3 years, achieving excellent sprint performance and delivery records.
- Resolved long-standing system issues, including memory leaks and critical 5xx errors due to ulimit.

Member of Technical Staff

06/2021 – 05/2023

- Developed an email blacklist feature that eliminated hard bounces, significantly improving key performance metrics (KPR).
- Extended SMS services to support Spanish, increasing accessibility for the US-based user base.
- Designed and developed a vendor driver library, enabling % based traffic migration with multi-vendor partitioning.
- Built a Two-Way SMS service, facilitating 100,000 patient replies per day for appointment confirmations, cancellations, or rescheduling.

Associate Member of Technical Staff

06/2019 – 05/2021

- Implemented a last-minute reminder system for patient appointments, reducing no-show rates by 18%.
- Resolved high-priority bugs impacting production systems, improving product stability and client satisfaction.
- Conducted Proof of Concepts to explore client needs and implemented key features to improve user engagement and workflow efficiency.
- Performed a PoC to explore the randomness of the results for enhancing a feature

TiVo, Bengaluru, India

06/2018 – 07/2019

Software Engineer Intern

- Enhanced the TV on-screen guide search experience, improving usability and content discoverability.
- Fixed long-standing bugs related to live content display, improving user experience.
- Developed features to preserve user state during navigation, ensuring a seamless transition when returning to previous screens.
- Added Dashboard support to the Set-Top Box Simulator, facilitating easier debugging of On Demand Content features using Microsoft Visual C++.

PROJECTS

- **Novel Generator using LLM Agentic Framework and RAG Pipeline:** Designed a tool that takes user-provided information, generates novel chapters using an LLM Agentic framework and a RAG pipeline, extracts metadata from the generated content, and stores it in a vector database for use in subsequent chapter generations.
- **Fine-Tuning Flan-T5 for Entity Extraction:** Fine-tuned the Flan-T5 model on a custom instruction-based entity extraction dataset using LoRA, improving extraction accuracy by approximately 25% compared to the original Flan-T5 model. [GitHub](#)
- **Organ Donor & Recipient Matching DApp:** Developed a decentralized application using Ethereum Solidity and Web 3.0 to match organ donors with recipients.

CERTIFICATIONS & ACHIEVEMENTS

- Certification in “Generative AI with Large Language Models”
- Awarded Starlight Award for 4 releases consecutively at athenahealth
- ConsenSys Certified Ethereum Blockchain Developer
- Winner, ConsenSys India Blockchain Hackathon 2018
- University Innovation Fellow, Stanford University
- Google Developer Student Clubs Lead
- Design Thinking workshop organizer for Google and Stanford University
- Android application workshop organizer for Google DSC program